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Installation of Node-Red in Windows

❑ Node.js Installation

Download the Node.js Installer

- Go to the official Node.js website: https://nodejs.org

- You will see two versions:

  - **LTS (Recommended for Most Users):** This is the Long Term Support version. It is the most stable and reliable. **You should download this one.**
  - **Current:** This is the latest version with the newest features, but it may be less stable. It's for users who need the latest updates.

- Click on the **LTS** button to download the Windows Installer (.msi file). The download will start automatically.

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Start typing... Ctrl + K

Download Node.js®

Get Node.js® v22.19.0 (LTS) for Windows using Docker with npm

Info Want new features sooner? Get the **latest Node.js version** instead and try the latest improvements!

```
# Docker has specific installation instructions for each operating system.
# Please refer to the official documentation at https://docker.com/get-started/

# Pull the Node.js Docker image:
docker pull node:22-alpine

# Create a Node.js container and start a Shell session:
docker run -it --rm --entrypoint sh node:22-alpine

# Verify the Node.js version:
node -v # Should print "v22.19.0".

# Verify npm version:
npm -v # Should print "10.9.3".
```

PowerShell Copy to clipboard

Docker is a containerization platform. If you encounter any issues please visit Docker's website ↗

Or get a prebuilt Node.js® for Windows running a x64 architecture.

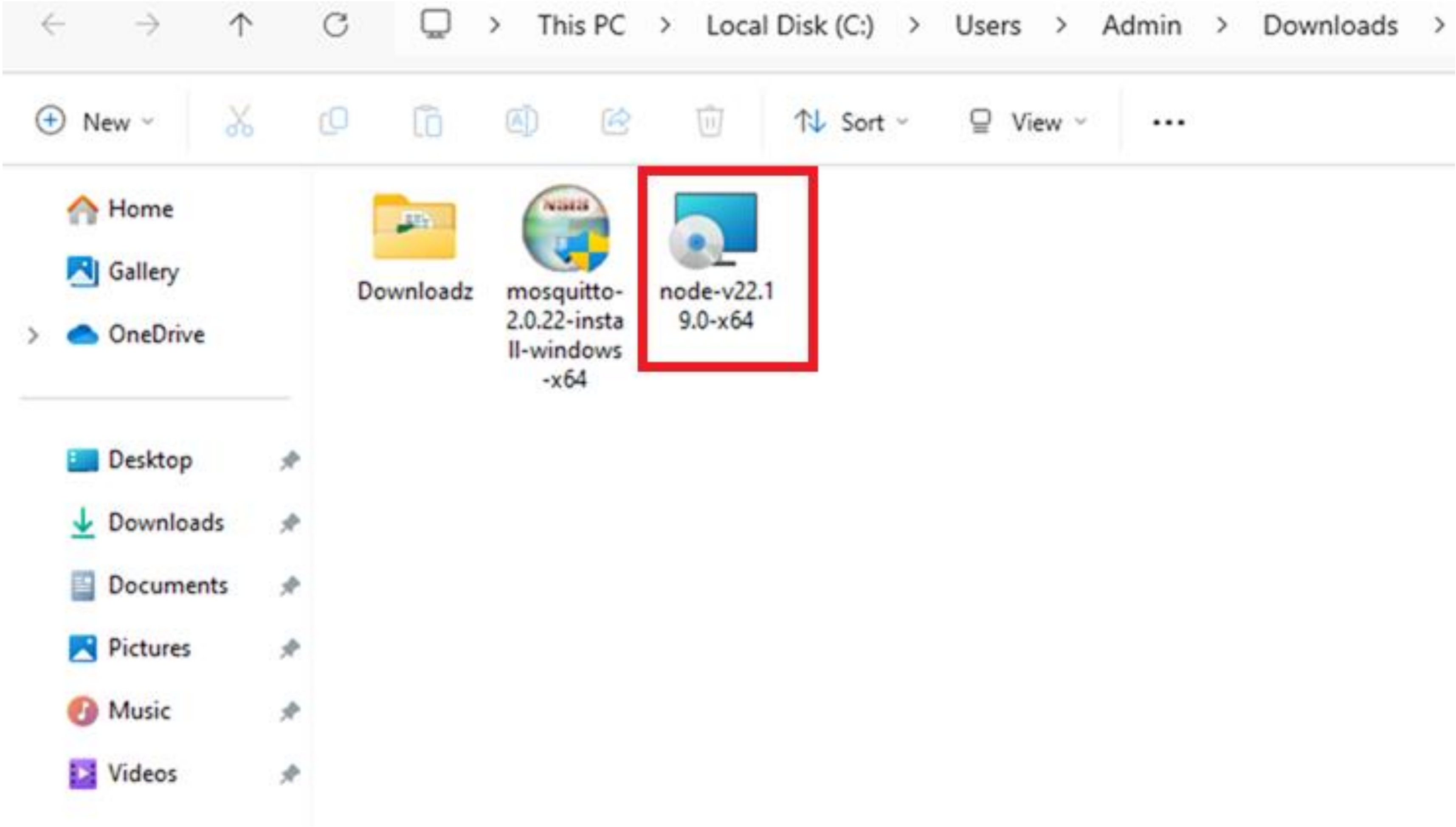
Windows Installer (.msi) **Standalone Binary (.zip)**

Run the Installer

6. Click the **Install** button. You may see a User Account Control (UAC) prompt; click **Yes**.
7. Wait for the installation to complete. Click **Finish**.

Run the Installer

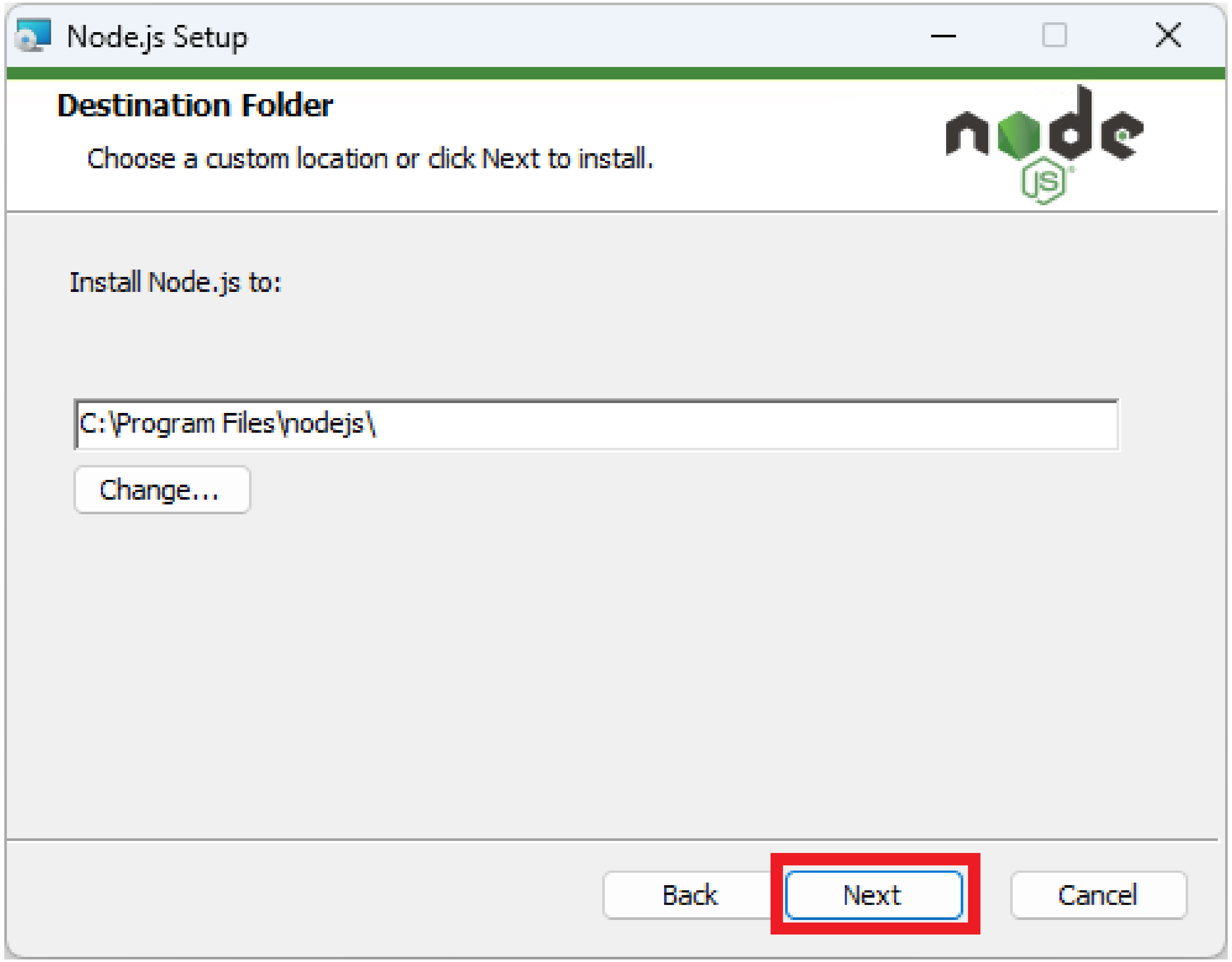
- Locate the downloaded file (e.g., node-v18.17.1-x64.msi) and double-click to run it.



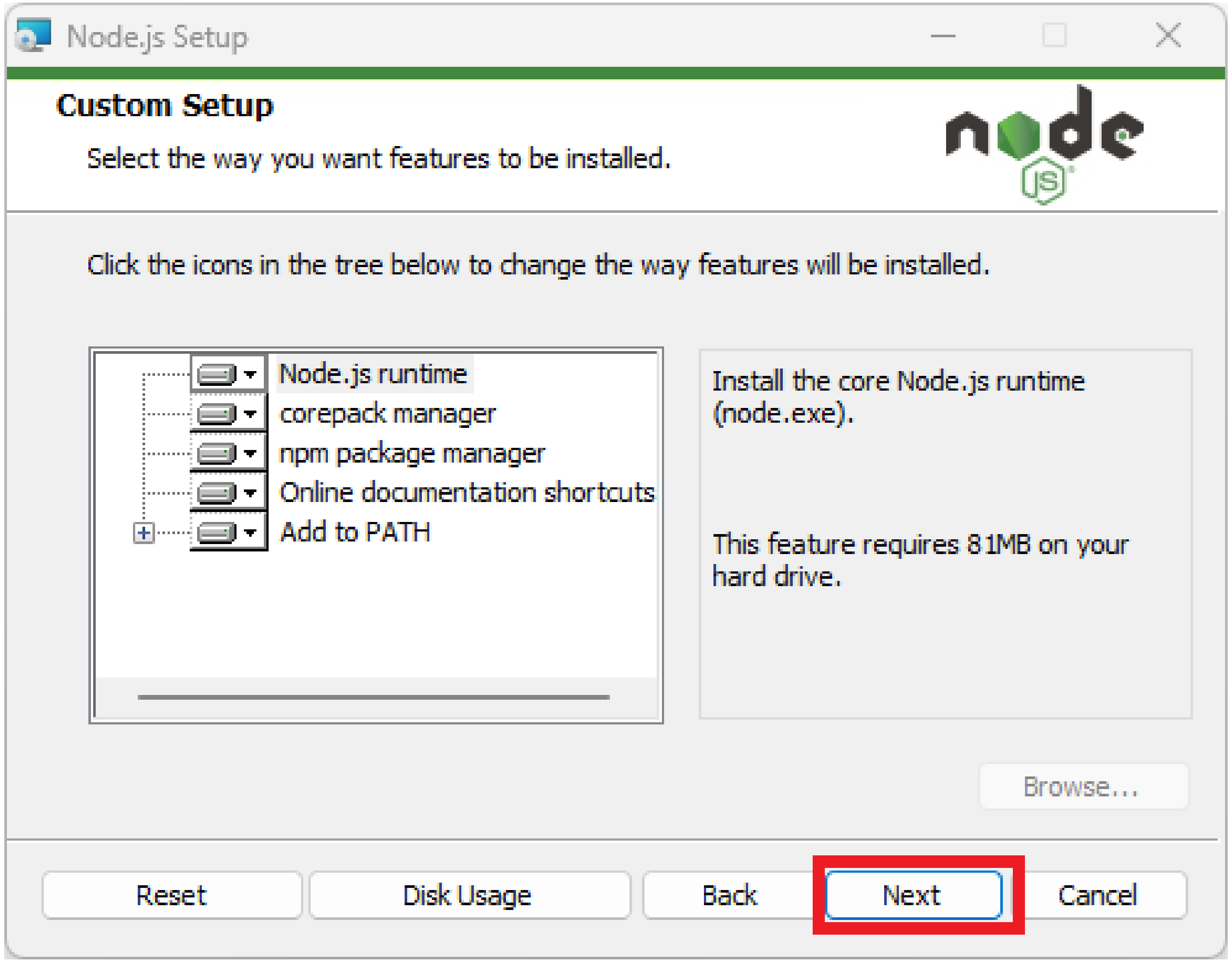
- The Node.js Setup Wizard will open. Click **Next**.

- **Accept the license agreement** and click **Next**.

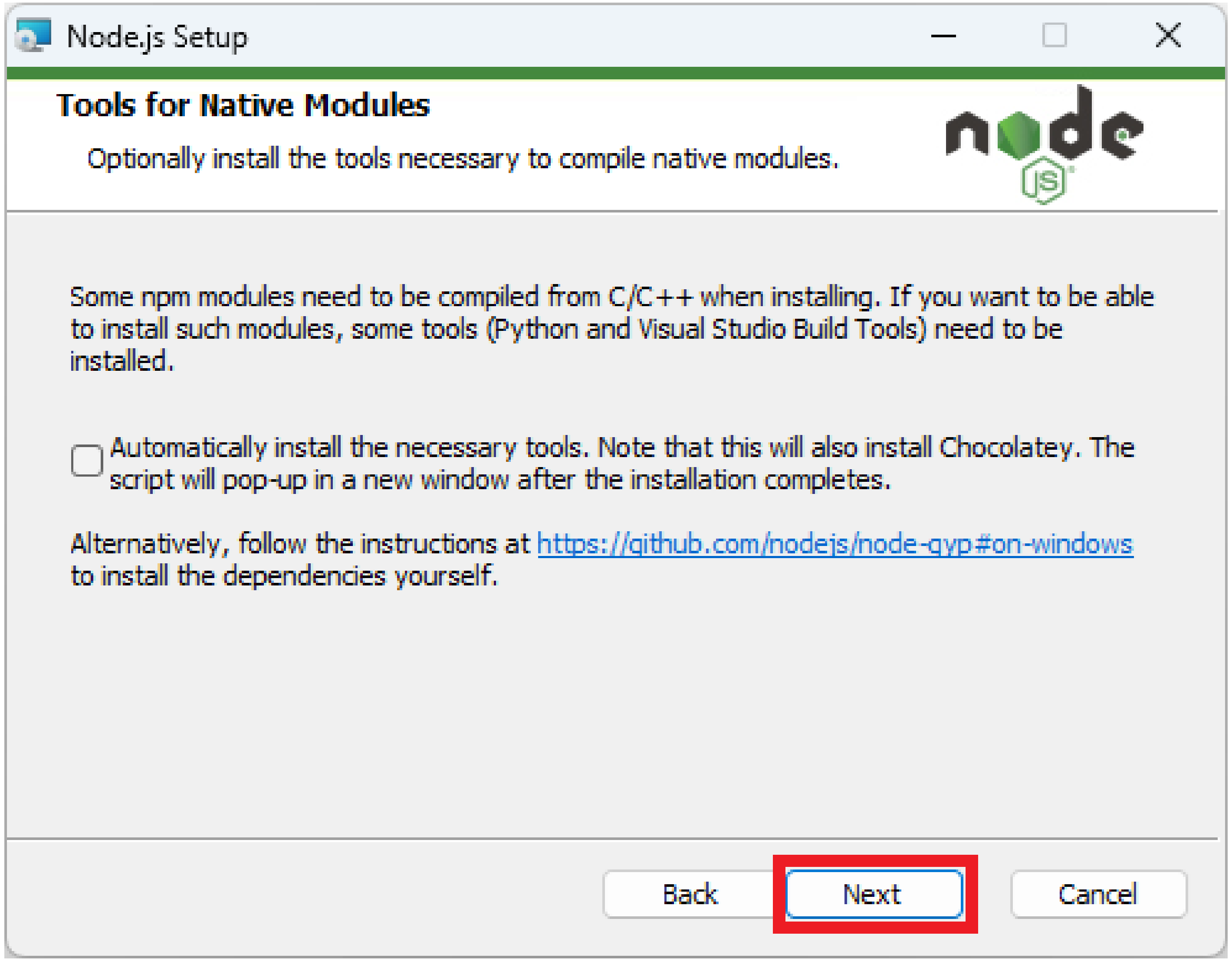
- **Choose the Installation Folder:** The default location is fine for most users (C:\Program Files\nodejs\). Click **Next**.



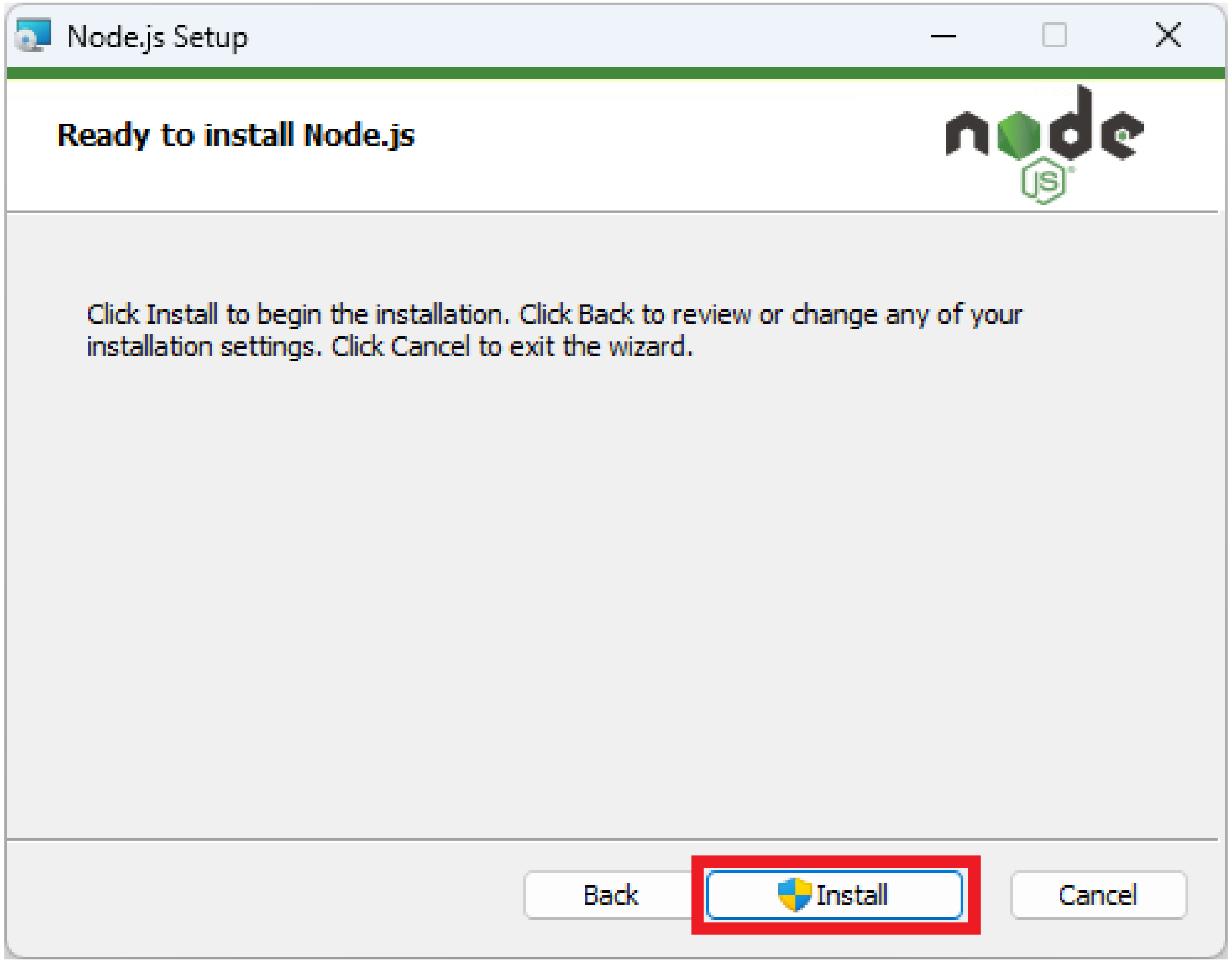
- **Customize the Setup:** The default settings on this page are perfect. It will:
    - Install Node.js core.
    - Install **npm** (the Node Package Manager, which is essential).
    - Automatically add Node.js and npm to your system **PATH**.
    - Download necessary tools (this option is fine to skip). Click **Next**.


Node.js Setup
Custom Setup
Select the way you want features to be installed.
Click the icons in the tree below to change the way features will be installed.
Node.js runtime
corepack manager
npm package manager
Online documentation shortcuts
Add to PATH
Install the core Node.js runtime (node.exe).
This feature requires 81MB on your hard drive.
Browse...
Reset
Disk Usage
Back
Next
Cancel

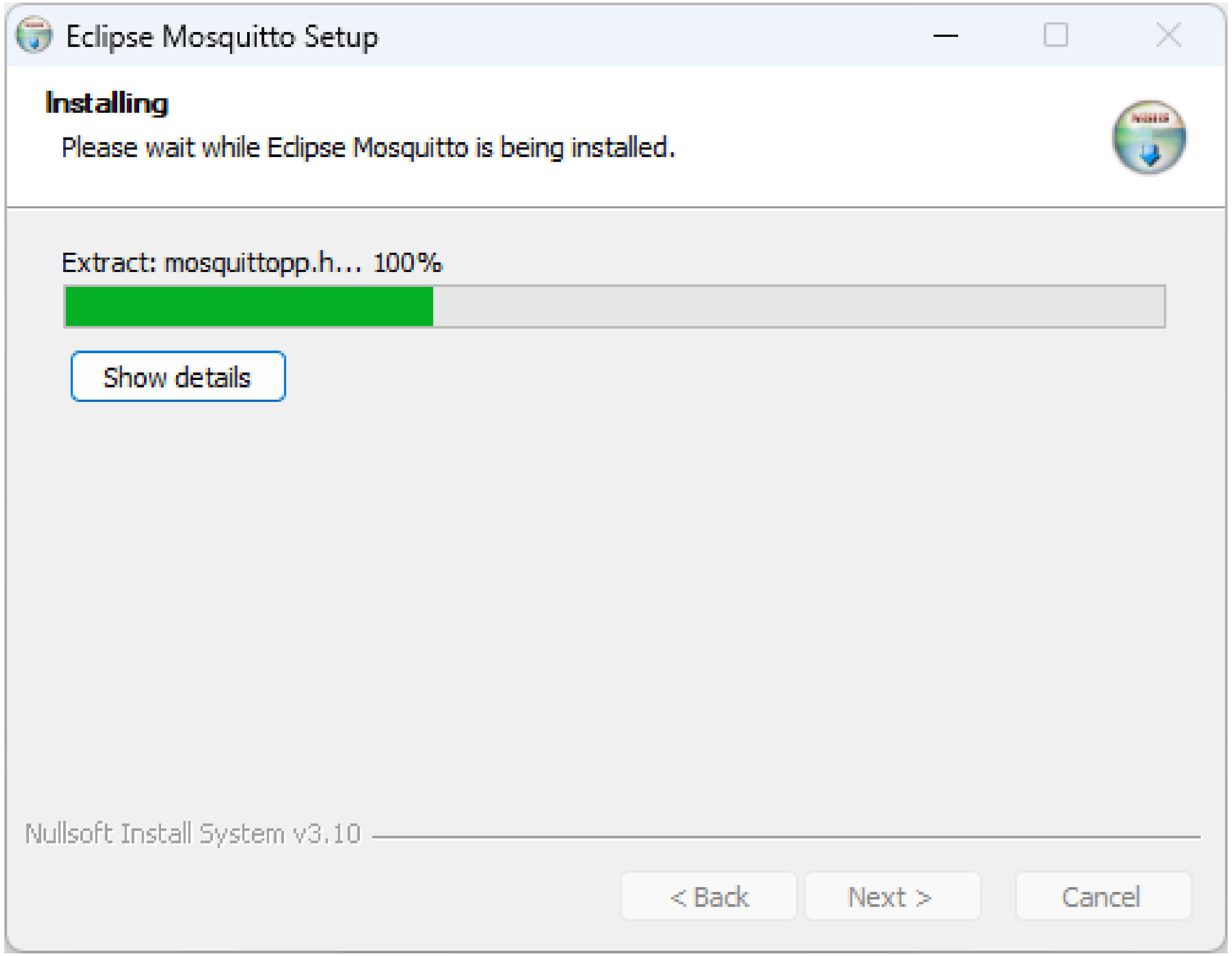
Select next. Don’t click on the checkbox



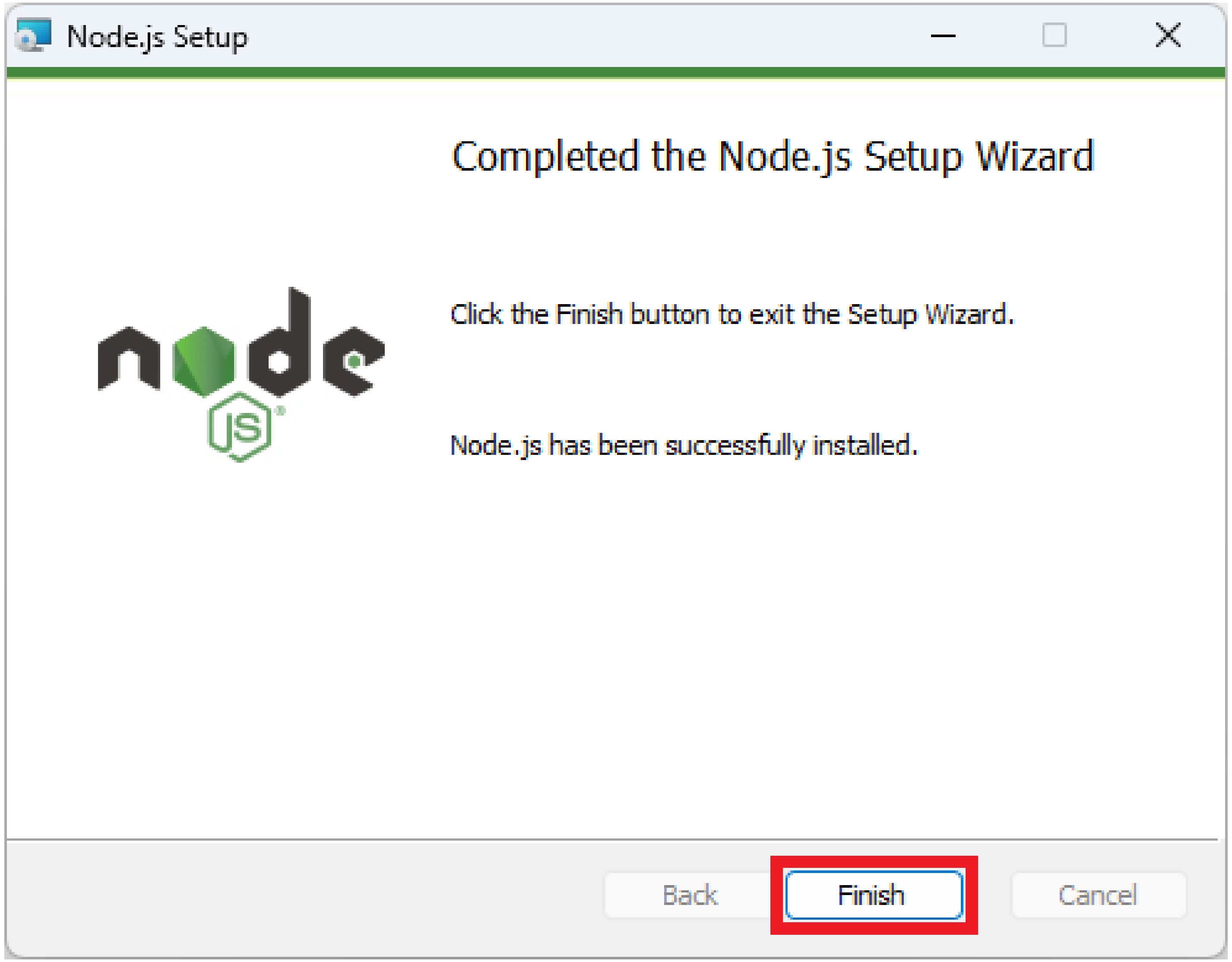
- Click the **Install** button. You may see a User Account Control (UAC) prompt; click **Yes**.



The installation is in progress.



Wait for the installation to complete. Click **Finish**.



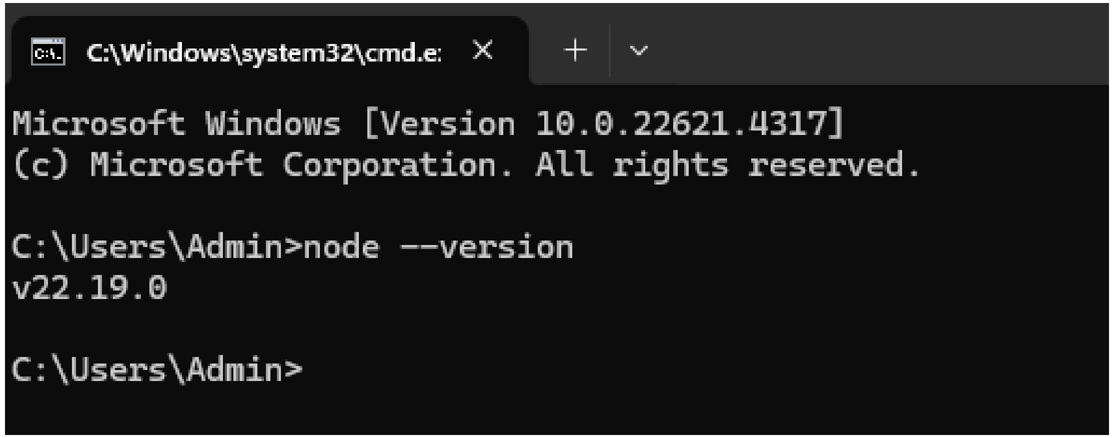
Verify the Installation

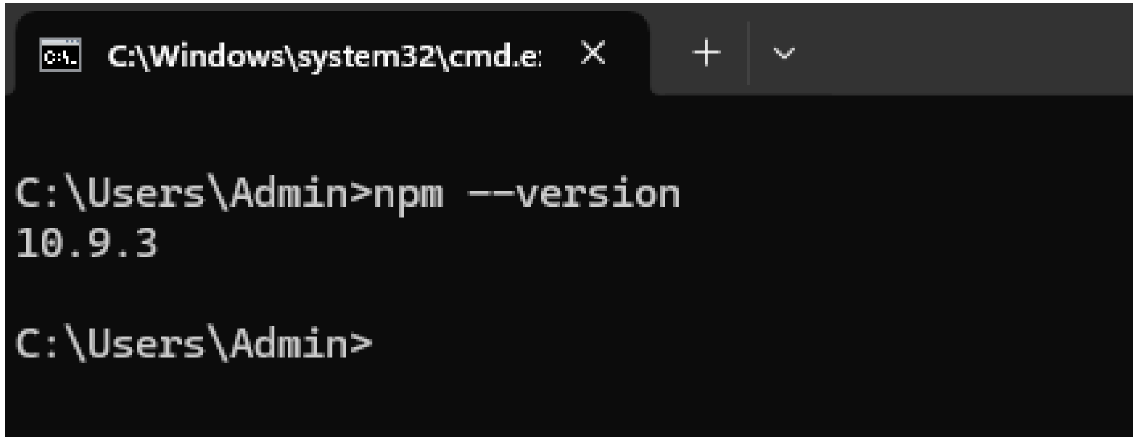
- Open the **Command Prompt**:

  - Press Win + R, type cmd, and press Enter.
  - Or, type "cmd" in the Windows Search bar and click on the Command Prompt app.

- In the Command Prompt window, type the following command and press Enter:

node –version
npm --vedsion





❑ Node-Red Installation

- Now, it is required to install Node-RED for Windows. Please go to the following site: https://nodered.org/docs/getting-started/windows

▪ Install Node-RED Globally

In the same Command Prompt window, type the following command and press Enter:

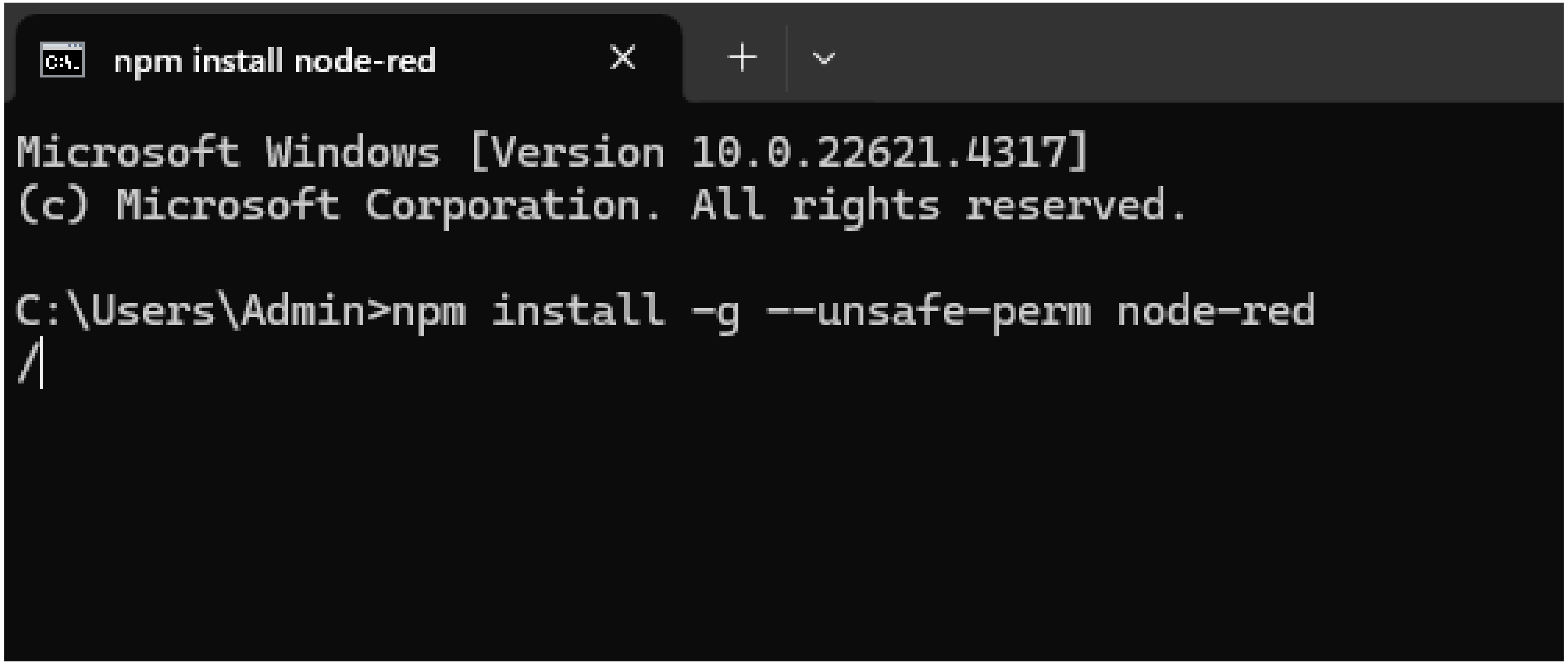
npm install -g --unsafe-perm node-red

- npm is the Node Package Manager, which you just installed.
- install tells npm to install a package.
- -g means "global", making the node-red command available from any folder on your system.
- --unsafe-perm is a flag often needed on Windows to avoid permission issues during installation.

- **Wait for the installation to complete.** npm will download and install Node-RED and all its dependencies.

- This may take a few minutes. You will see a lot of text scrolling by.

Installation is in progress.



```
C:\Windows\system32\cmd.e:

Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>npm install -g --unsafe-perm node-red

added 276 packages in 2m

52 packages are looking for funding
  run `npm fund` for details
npm notice
npm notice New major version of npm available! 10.9.3 -> 11.6.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.6.0
npm notice To update run: npm install -g npm@11.6.0
npm notice

C:\Users\Admin>
```

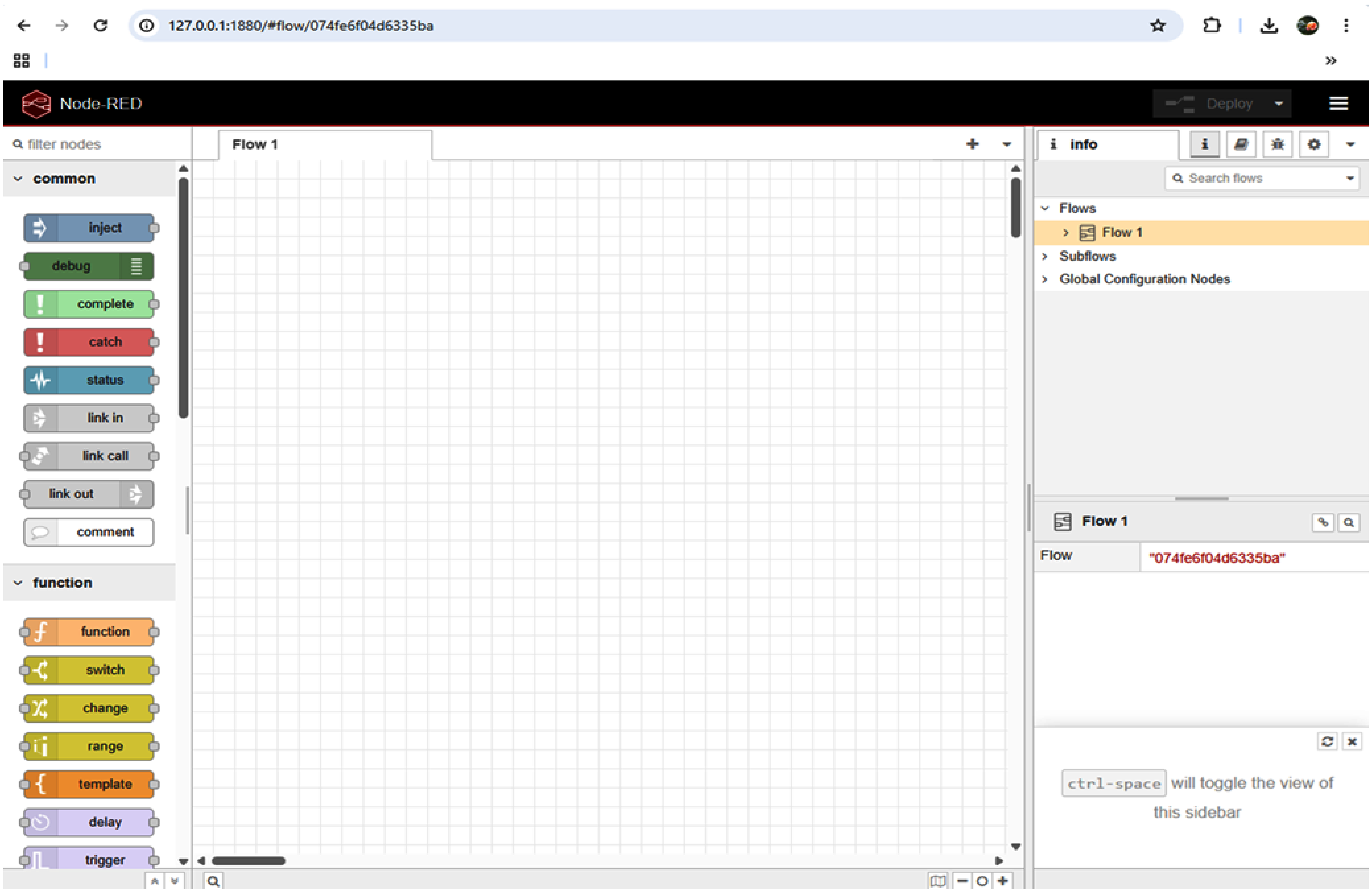
- Once the installation is complete, close the Command Prompt.
- Then, open a new Command Prompt, type node-red, and press Enter.
- You will see a welcome message displayed

```
node-red

Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>node-red
5 Sep 22:31:14 - [info]

Welcome to Node-RED
===================

5 Sep 22:31:14 - [info] Node-RED version: v4.1.0
5 Sep 22:31:14 - [info] Node.js  version: v22.19.0
5 Sep 22:31:14 - [info] Windows_NT 10.0.22621 x64 LE
5 Sep 22:31:14 - [info] Loading palette nodes
5 Sep 22:31:15 - [info] Settings file  : C:\Users\Admin\.node-red\settings.js
5 Sep 22:31:15 - [info] Context store  : 'default' [module=memory]
5 Sep 22:31:15 - [info] User directory : \Users\Admin\.node-red
5 Sep 22:31:15 - [warn] Projects disabled : editorTheme.projects.enabled=false
5 Sep 22:31:15 - [info] Flows file     : \Users\Admin\.node-red\flows.json
5 Sep 22:31:15 - [info] Creating new flow file
5 Sep 22:31:15 - [warn]
5 Sep 22:31:15 - [warn] Encrypted credentials not found
5 Sep 22:31:15 - [info] Server now running at http://127.0.0.1:1880/
5 Sep 22:31:15 - [info] Starting flows
5 Sep 22:31:15 - [info] Started flows
```

- http://127.0.0.1:1880/

- You will see a web page address displayed—copy it and paste it into your default browser.

- The Node-RED dashboard will appear, and it is now ready to use.


127.0.0.1:1880/#flow/074fe6f04d6335ba
Node-RED
Deploy
filter nodes
common
inject
debug
complete
catch
status
link in
link call
link out
comment
function
function
switch
change
range
template
delay
trigger
Flow 1
info
Search flows
Flows
Flow 1
Subflows
Global Configuration Nodes
Flow 1
Flow
"074fe6f04d6335ba"
ctrl-space will toggle the view of this sidebar

Add Dashboard and MQTT Nodes:

- To create the user interface, you'll need the Node-RED Dashboard.
- In the Node-RED editor, click the **hamburger menu (☰)** in the top right, select **Manage palette**, go to the **Install** tab, and search for **node-red-dashboard** and **node-red-contrib-mqtt-broker**. Install both.

Create the Flow:

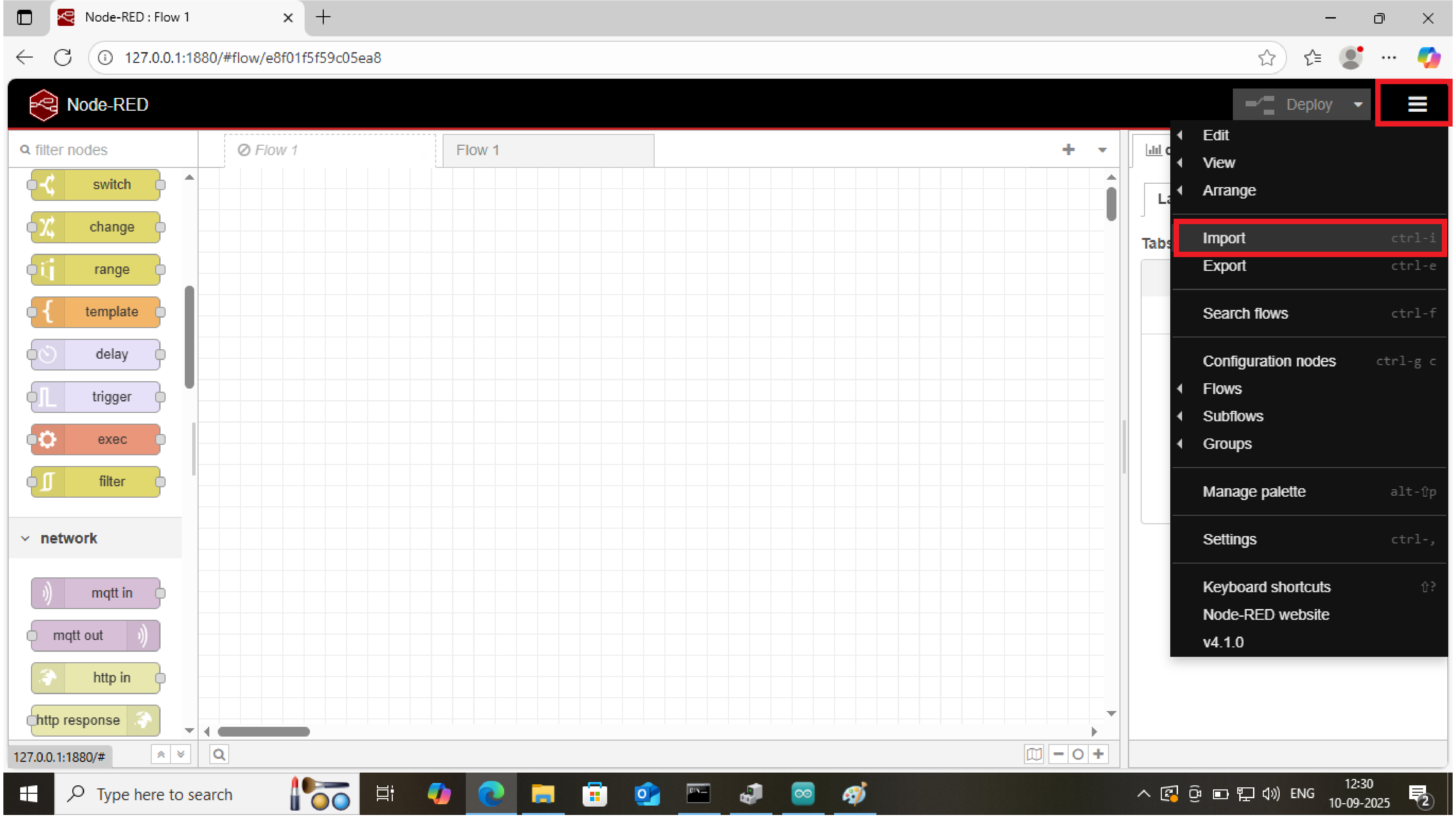
- Drag an **mqtt out** node from the palette onto the canvas. This node will publish messages to the MQTT broker.
- Double-click the **mqtt out** node. For the "Broker," click the edit pencil and configure it. Set the server to your computer's IP address (the same one you put in the ESP32 code) and the port to **1883**. The "Topic" should be esp32/led, matching what the ESP32 is subscribed to.
- Drag a **button** node from the dashboard section onto the canvas. This will be your "ON" switch. Double-click it and set the "Label" to "ON" and the "Payload" to **"ON"**.
- Drag a second **button** node. Configure it as the "OFF" switch with the "Label" as "OFF" and the "Payload" as **"OFF"**.
- Connect the output of both button nodes to the input of the **mqtt out** node.

Execution

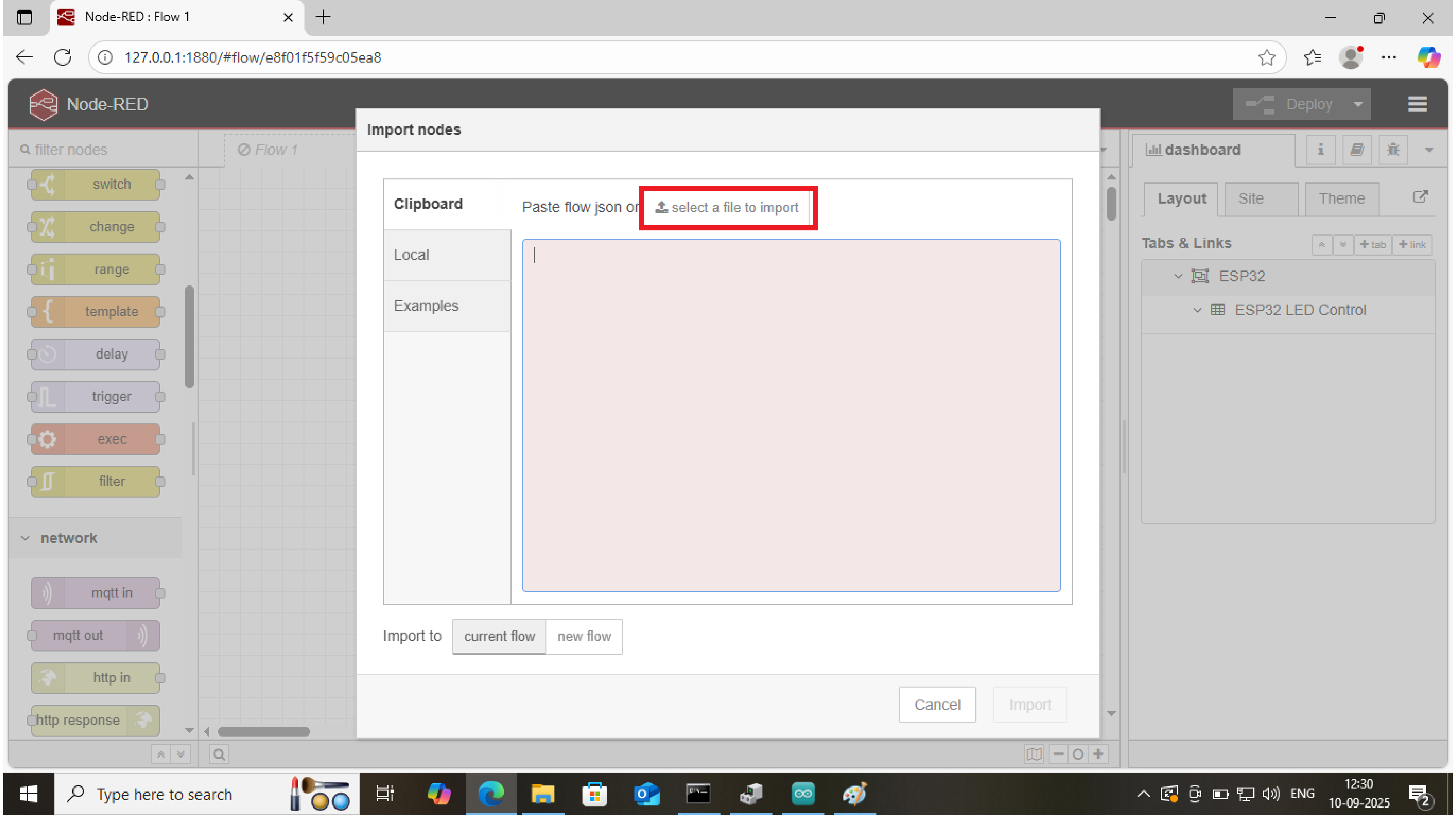
Now, you can test the setup.

- Ensure the ESP32 is powered on and connected to your Wi-Fi network.
- Start Node-RED from your terminal with the command node-red.
- Open the Node-RED dashboard by navigating to http://YOUR_COMPUTER_IP:1880/ui in your web browser. You will see the two buttons.
- Click the **ON** button, and the LED on your ESP32 should turn on. Click the **OFF** button, and it should turn off. This demonstrates the end-to-end communication from your web browser through Node-RED and the MQTT broker to your ESP32.

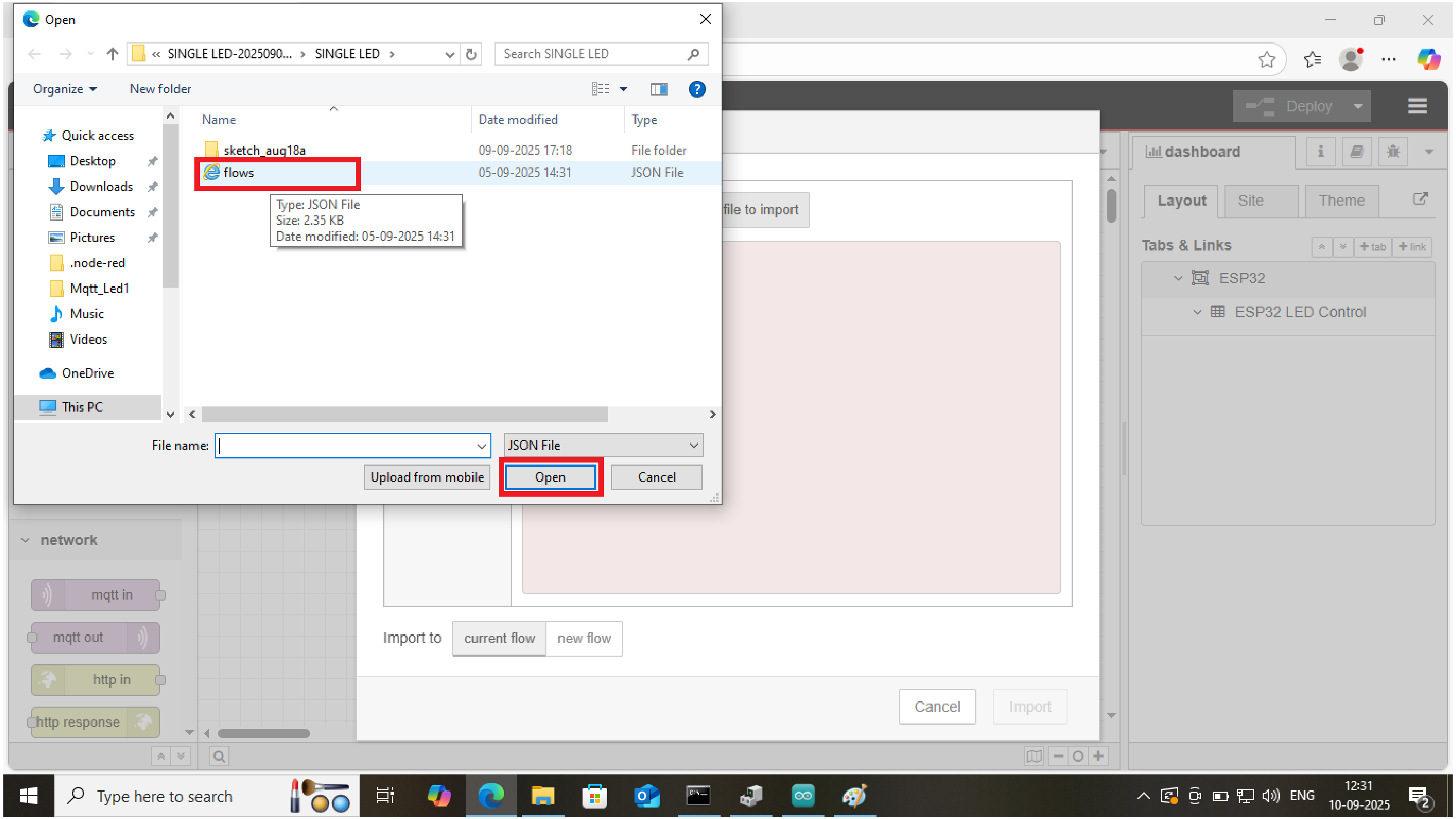
In the Node-RED editor, click the hamburger menu (☰) in the top-right corner and select 'Import



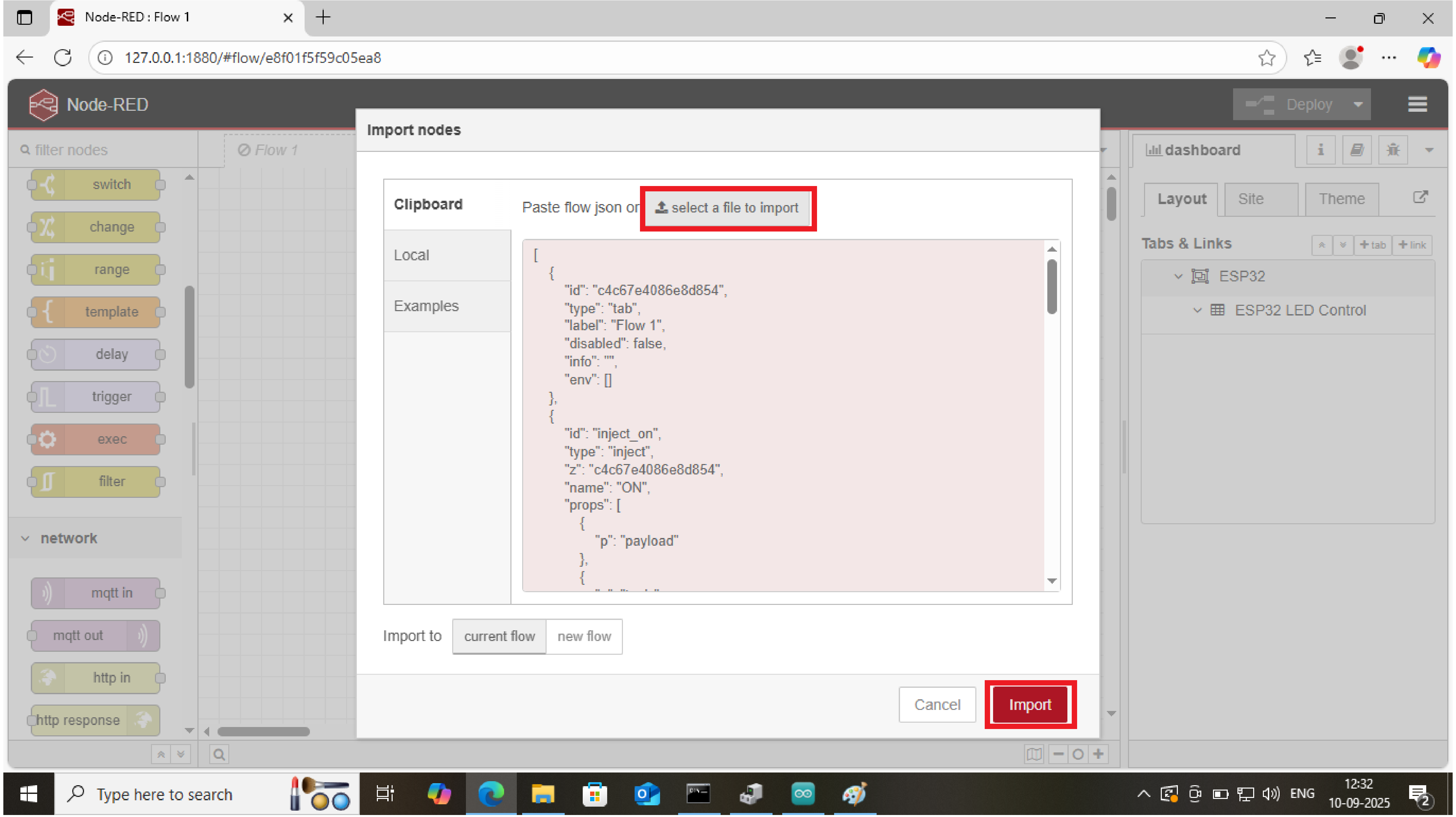
After selecting the 'Import' option, select the file you wish to import.



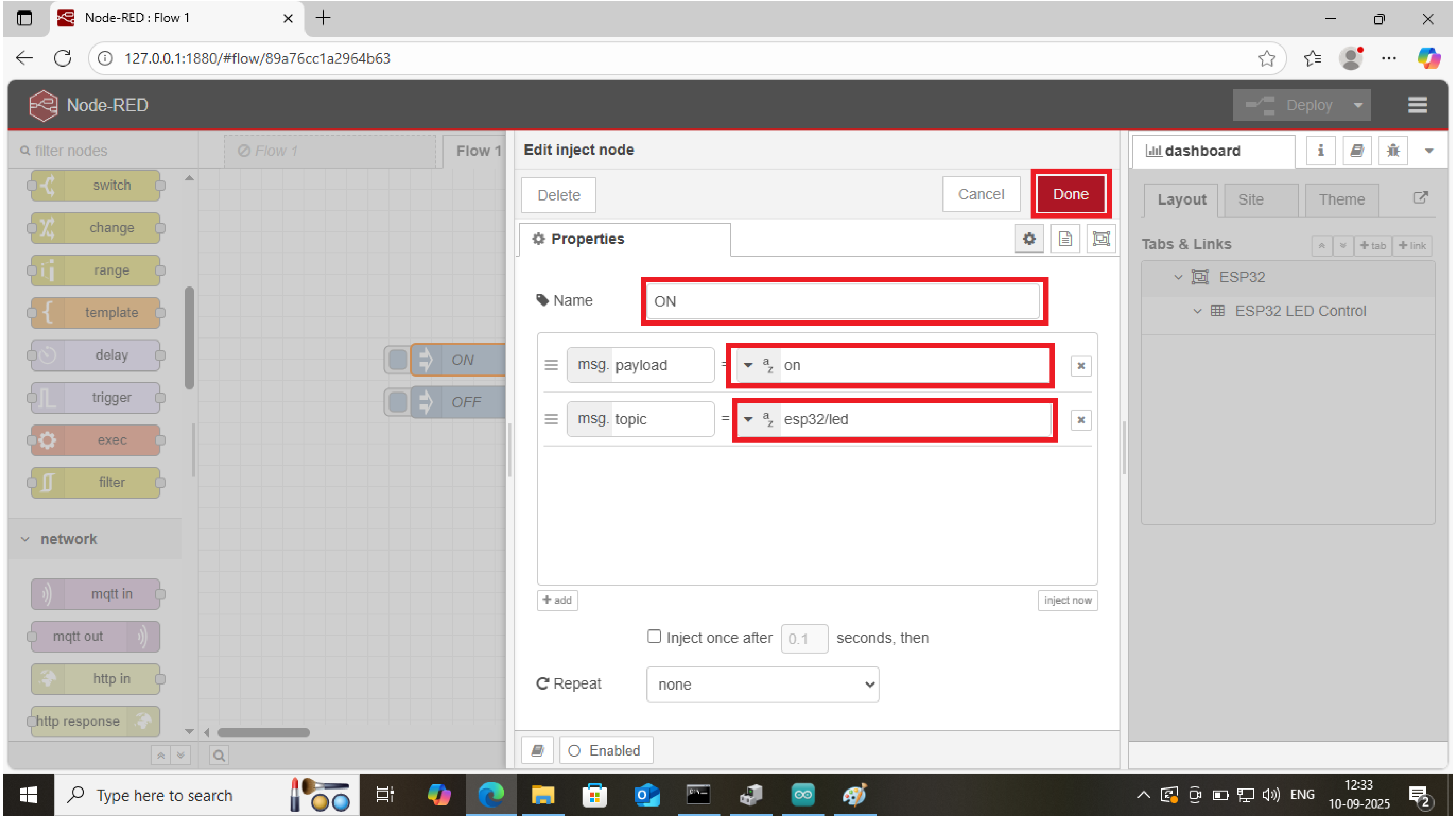
After choosing the 'Import' option, browse to the flow file and select 'Open' to proceed.



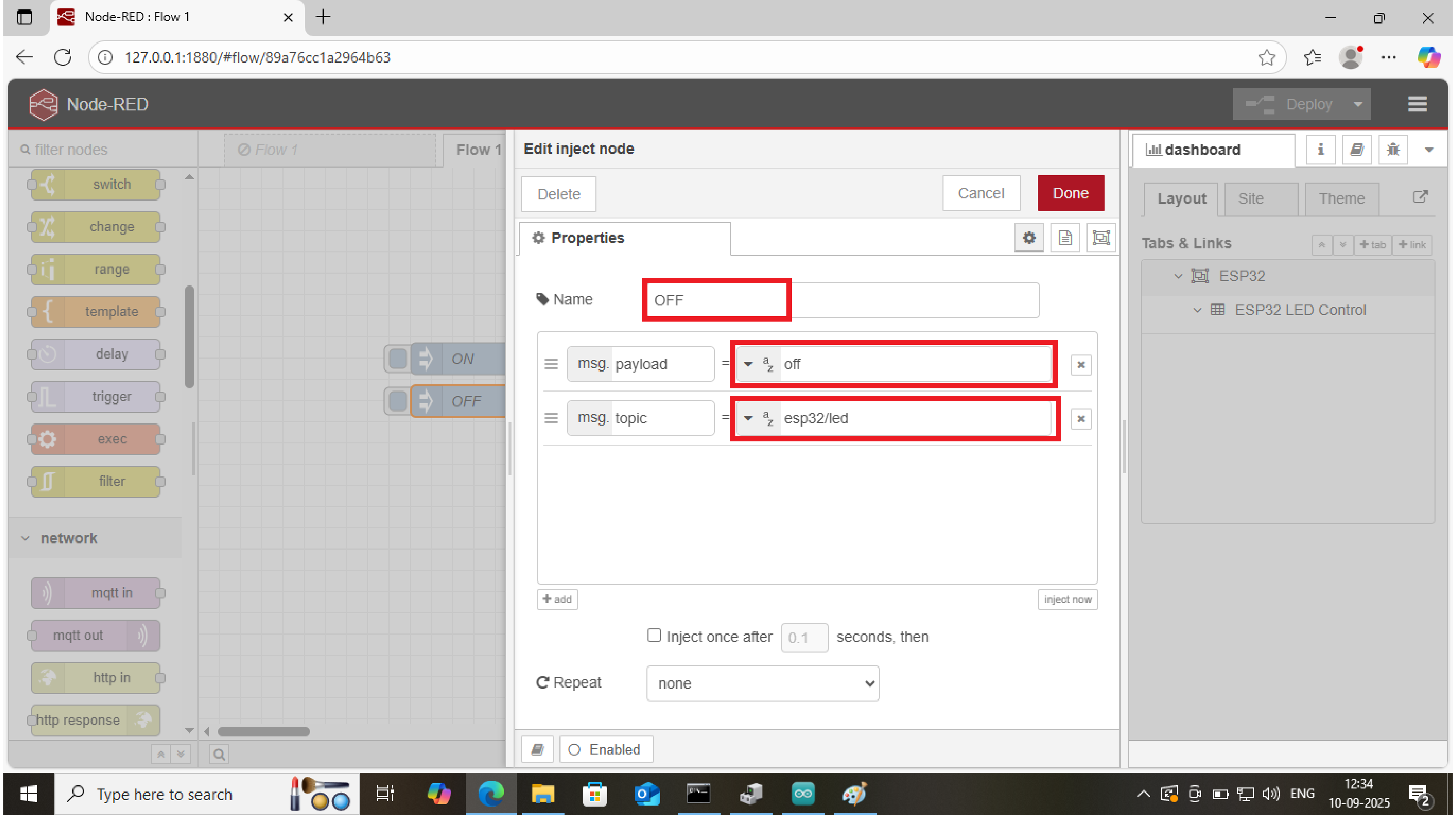
Once the file is open, click on the 'Import' option.



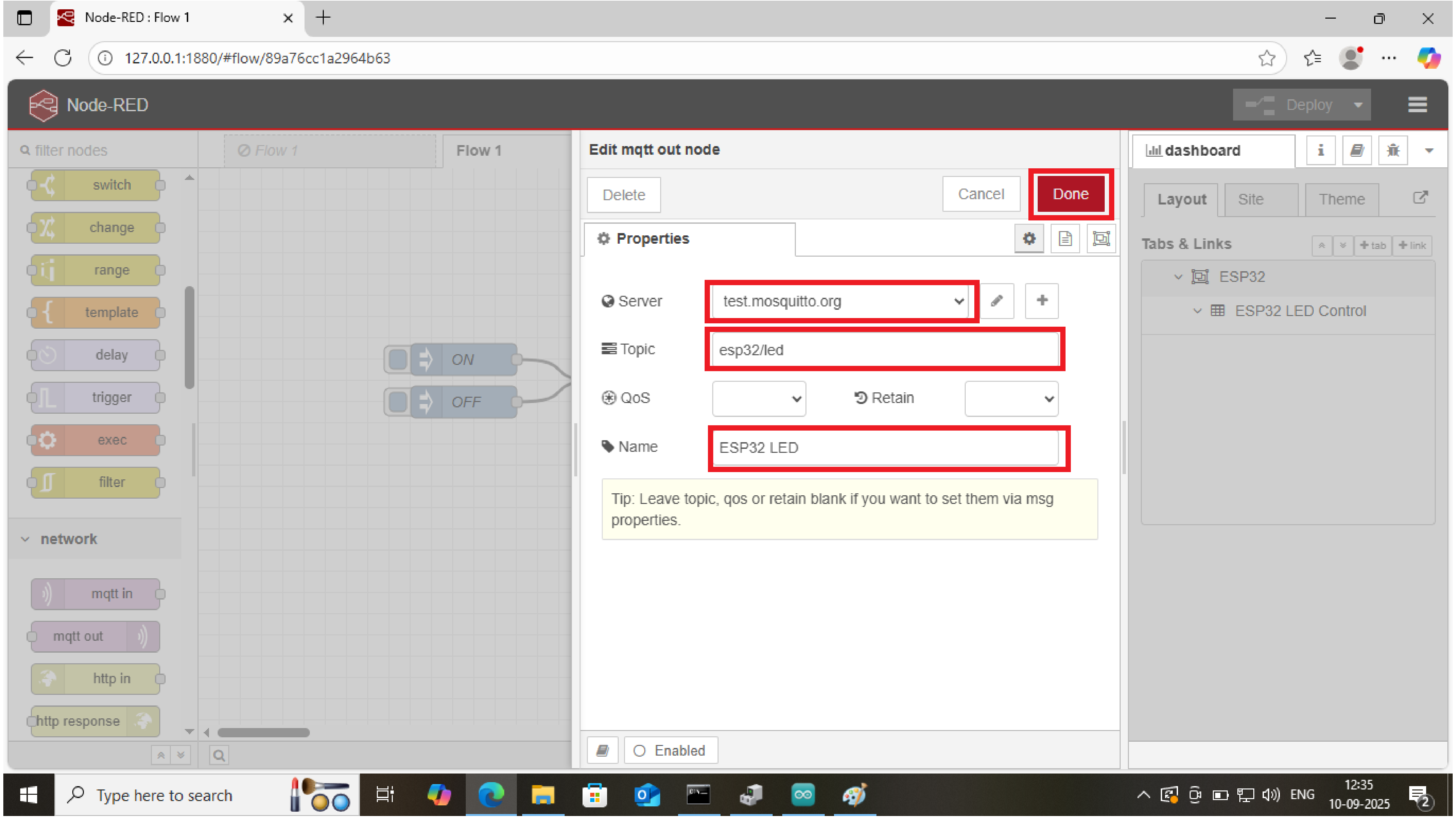
Next, double-click the first node to edit it, input the necessary settings, and select 'Done'.



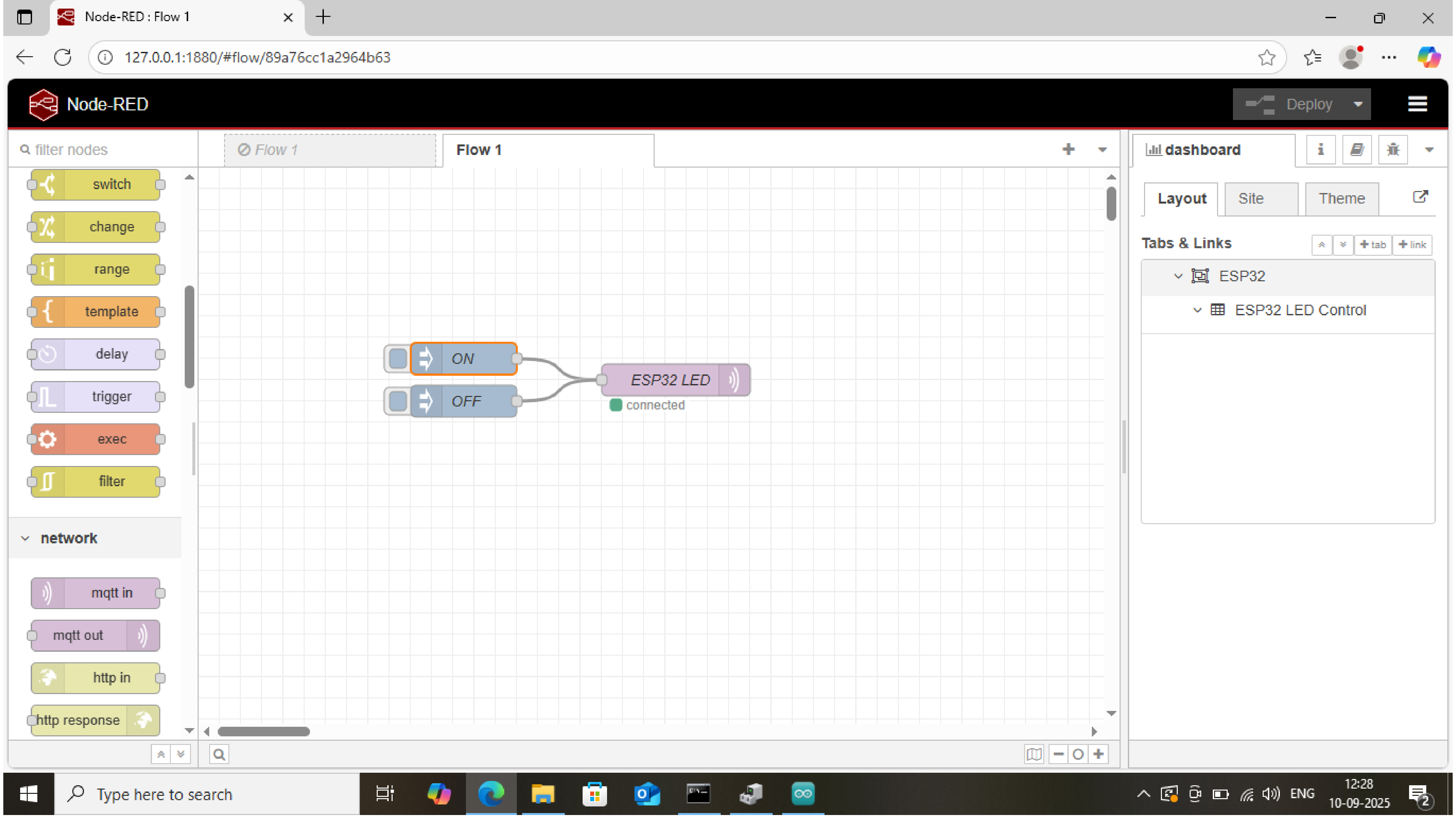
Next, double-click the second node to edit it, input the necessary configuration, and select 'Done' to save the changes.



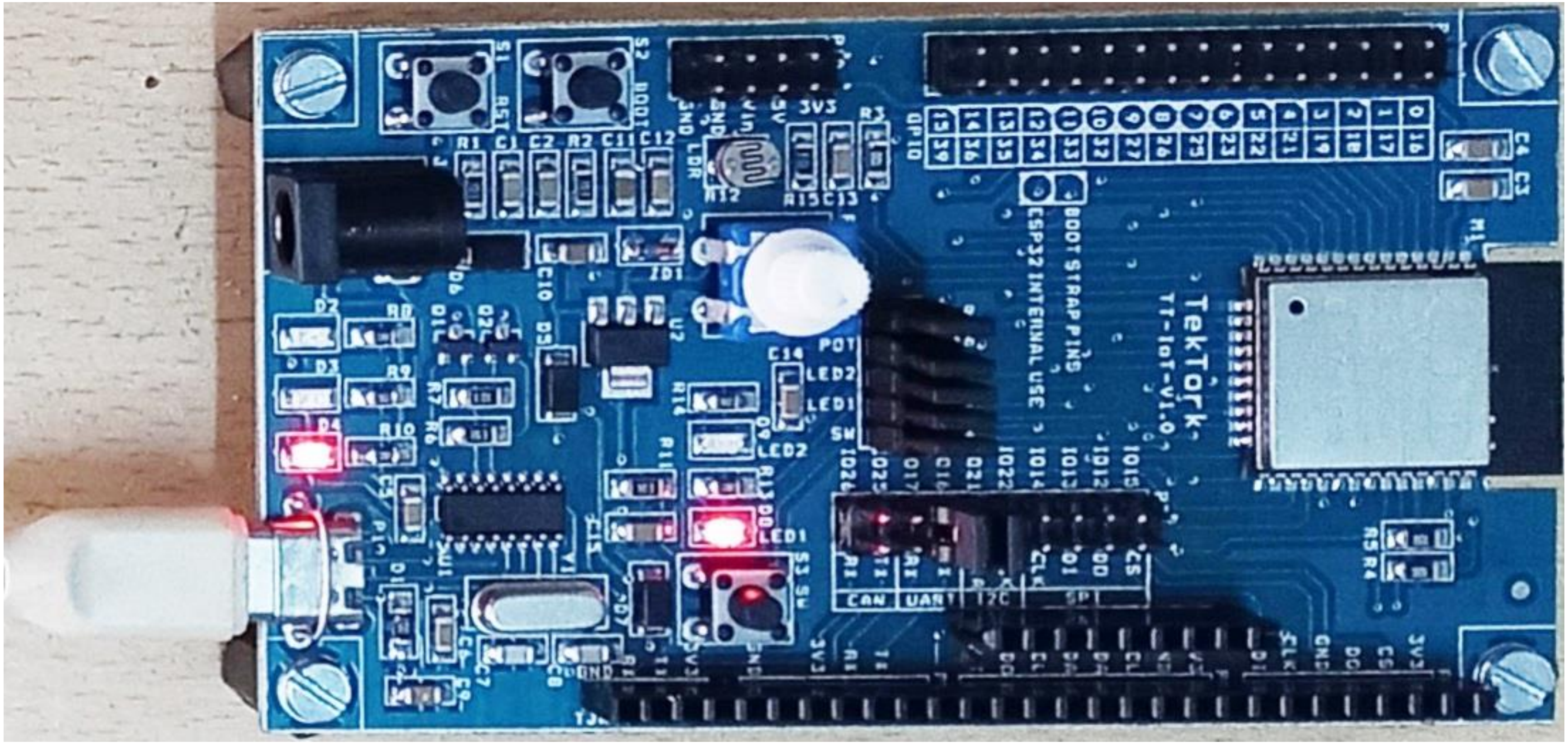
Next, double-click the third node (mqtt)to edit it, provide the necessary configuration, and click 'Done' to save your changes



Finally, click the 'Deploy' button to connect the nodes to the MQTT server. You can now control the LED on the ESP32 using the Node-RED flow



Output



Thank You